Mobile AMR integrated control system

Interface protocol

V3.2.0

Preface

Purpose

This document is written to establish interface protocol implementation specifications between third-party systems and ICS systems. Subsequent customer docking development will be based on this document.

Reading object

Customer business responsible personnel, customer R&D technicians, and API callers.

Symbol conventions

The following symbols may appear in this document and their meanings are as follows:

Table 1 Symbol meaning

Logo	Description
说明	It indicates additional information to the main text, which emphasizes and supplements the main text.
窍门	It means it can help you solve a problem or save you time.

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Chapter 1 Technical Conventions

1.1 Technology used

- Both systems use JSON data format.
- Realize the interaction between the two system interfaces through Http Post protocol.
- All API calls adopt synchronous calling mode.
- Data is uniformly encoded and decoded using the utf-8 character set.
- The interactive data content must include the content specified in this document, and some non-mandatory fields may not be passed in after agreement.
- Developers from both parties develop interfaces based on the JSON samples and API interface description documents provided by the other party.
- The default port is usually 7000, the IP is the RCS platform server IP, and the actual interface request address and port depend on the on-site situation.

1.2 Definitions and explanations of some important values

1.2.1 Description of important values of request parameters

Table 1-1 Description of some important values of request parameters

Parameter name	Type	Maximum length	Description
orderId	string	100	Task order number: A task order number is generated for each transport task. During the transport process, both systems use this number to exchange data. Duplication is not allowed.
modelProcessCode	string	100	Business process name and number: A type of production business process name corresponds to a number. You can query the "Business Process Number" column on the "Business Process Template" page on the RCS client to obtain the corresponding number.
fromSystem	string	50	Source system: Defined by the third-party system and informed to the ICS system
areaId	string	10	Area Id: This field is mainly used to distinguish different map areas. You can obtain it by querying "Configuration Management> Area Management" on the RCS client. The default value of areaId is: 1
deviceNum	string	32	Equipment number: Each AMR will be manually assigned an equipment number, which can be obtained from the "Equipment Management" page of the RCS client.

1.2.2 Description of the basic structure of the return value

Table 1-2 Description of the basic structure of return value

Parameter name	Is it required?	Description
code	Required	Return code: 1000: Success Other: Failed
desc	Required	Returns encoding description
data	Optional	Returned data content

1.2.3 Result code definition

Table 1-3 Result code definition

Result code	Description
1000	Request successful
1001	Request failed
7201	Wrong request parameters
2103	The task flow template number does not exist
2107	The paths collection is empty
2108	The path collection is incorrect
2111	Please do not issue tasks repeatedly.
2115	Mission accomplished
Other	Other errors

1.3 Authentication request instructions

1.3.1 Brief description of AMR interface authentication method

- The AGV interface uses WSSE authentication method.
- This authentication is closed by default in the intranet, and third parties can use the simplest http+json request.
- In the public network, authentication needs to be enabled to perform necessary security checks and data interception. This situation requires that third-party interface requests must implement authentication logic. Use the username and password provided by the AMR system to make requests.

1.3.2 WSSE authentication encryption code logic

- Obtain authorization from the person in charge of the AMR system and obtain the username userName and password password.

- Get the timestamp in seconds stTime.
- Get the Str string of timestamp + password.
- Convert the string encrypted by SHA-1 of Str to Base64 and encode it to get a digest token.
- Assemble a WSSE string in the format of: "useName + ":" + stTime + ":" + digest" (an example of the assembled WSSE encrypted string is: admin:1623993204:rlqQDuh38bIj4SeyG1t5t+mh8lw=)
- Put the WSSE string into the Authorization in the HTTP request header.
- Make normal interface requests.

Chapter 2 Commonly used interfaces

Table 2-1 Brief description of commonly used interfaces of ICS system

Interface description	Detailed description	Notes
Task dispatching interface	The third-party system requests the ICS system to issue tasks.	Must be implemented
Continue task interface	The AMR business template configuration has a pause waiting for confirmation in a certain subtask process. When AMR executes this subtask, it will pause and wait for confirmation. After waiting for confirmation, the third-party system calls the interface to enable AMR to continue to execute subsequent tasks.	
Cancel task interface	Called when a third-party system cancels a task.	

Table 2-2 List of third-party system interfaces

Interface description	Detailed description	Notes
Task status receiving interface	It is called when the ICS system proactively reports the task status sent by the third-party system.	Must be implemented

2.1 Interfaces provided by the ICS system

2.1.1 Task issuance

2.1.1.1 Interface details

- Interface Name:http://IP:7000/ics/taskOrder/addTask
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description:
  - ◇ This interface is a general interface for the third-party system to send tasks to AMR.
  - ◇ Path point names support customization by third-party systems.
  - ◇ The integrated control system can be converted into AMR available points through mapping relationships.
- Usage scenario: Distribution of all tasks

2.1.1.2 Parameter details

Table 2-3 Request parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	modelProcessCode	string	100	Yes	Business process name and number. One type of production business process name corresponds to a number. You can query the "Business Process Number" on the "Business Process Template" page on the RCS third-party terminal to obtain the corresponding number.
	priority	int	1	No	Priority: 4: High 6: Medium 8: Low High priority tasks will be scheduled first, the default is 6
	fromSystem	string	50	Yes	Source system: The third-party system that issues tasks, customized by the third-party system. Such as: MES, WMS, etc.
	orderId	string	100	Yes	Task order number: This needs to be specified when a third-party system issues a task, and it must be ensured that each task order number is

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						unique and does not repeat the previous one. **说明** When issuing wave tasks, this order number is the wave order number.
	taskOrderDetail	subOrderId	string	100	No	● If the business process template contains multiple RCS tasks and you want to monitor the execution progress of each subtask, you can use this field. ● For wave tasks, the single business process task ID of the wave; required when issuing wave tasks
		taskPath	string	200	No	Task path point set: the starting point and target point of AMR operation; multiple points

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						are separated by English commas. The path point set supports the input of the patch area number maintained by RCS. After input, RCS is responsible for selecting available points from the input patch area (RCS needs to maintain the inventory location information of each point in the patch area)
		shelfNumber	string	32	No	Vehicle Number: ● Latent lifting AMR task, assigning shelf number ● Material box AMR task, assign material box code number ● For other AMR type tasks, the assignment is invalid.
		shelfModel	int	2	No	Vehicle type: 1: Square shelves 2: Rectangular shelves

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						3: Traction vehicle 4: Pallet 7: Forklift material box ● For material box AMR tasks, when AMR needs to identify the material box number, the material box model needs to be sent down for identification. ● The vehicle ID can be queried on the RCS client
		assignRobotIds	string	64	No	The set of candidate devices that can perform this task, pass in the "device serial number" and "device number", supports passing in multiple devices, and multiple devices are separated by English commas.

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
		robotType	string	64	No	Device models that support this task. Multiple models are supported, separated by commas.
		deviceNum	string	32	No	Specify a device to perform the task and pass in the "device number". Only one AMR device number is allowed and it must be used with releaseRobot (without releasing the vehicle function). The device number can be obtained through the task status reporting interface.
		releaseRobot	int	1	No	Whether to release after this task is completed: 1: Do not release AMR when the task is completed Other: Release AMR If not released, the vehicle can be assigned to perform subsequent tasks.
		extraInfo	Json	400	No	Additional fields
Answer	code		int	6	Yes	Return status code: 1000: Success

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					Other: Failed
	data	string	200	Yes	Return result content
	desc	string	64	Yes	Status code description

2.1.1.3 Examples

Normal task parameter request (pass in necessary parameters as needed, and do not pass in unnecessary parameters):

```
{
            "modelProcessCode": "chuku",
"priority": 6,
"fromSystem":"MES",
"orderId":"3215435",
"taskOrderDetail": [
                {
"taskPath":"Shelf area, workbench, shelf area",
                        "shelfNumber":"80000032",
                        "extraInfo":{"skuNo":"321d321","num":"3"}
                }
            ]
}
```

Wave task parameter request (pass in necessary parameters as needed, and do not pass in unnecessary parameters):

```
{
      "modelProcessCode": "batchTask",
"fromSystem": "DWCS1",
      "orderId": "wave20241023092040",
"priority": 4,
"taskOrderDetail": [
            {
                  "subOrderId": "5847541",
                  "taskPath":"20002,34001007",
                  "shelfNumber":"1112"
            },
              {
                  "subOrderId": "3547844",
                  "taskPath":"20004,34001007",
                  "shelfNumber":"1113"
            }
      ]
}
```

Successful response:

```
{
```

```
"code": 1000,
"data": "3215435",
"desc": "Request successful"
}
```

Response when task flow number is not configured:

```
{
"code": 2103,
"data": "",
"desc": "Task process template number does not exist"
}
```

说明

The common problem that returns this error code is that the business process template number is not configured. This configuration item is on the leftmost tab page of the task template configuration.

Response when the task number is issued repeatedly:

```
 {
"code": 2111,
"data": "",
"desc": "Please do not issue tasks repeatedly"
}
```

说明

The common problem reported for this error code is that the task has been sent to ICS, ICS has written it into the database, but the task with the same order number is sent repeatedly.

Response when there is an error in sending the task path:

```
{
"code": 2108,
"data": "",
"desc": "The number of mission point collections is incorrect"
}
```

说明

The common problem reported for this error code is that the number of points required to be sent by the RCS task template is inconsistent with the number of points actually sent.

2.1.2 Continue task*

2.1.2.1 Interface details

- Interface Name:http://IP:7000/ics/out/task/continueTask

- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- This interface is used when a task is paused at a certain step and a third-party system is required to confirm that the task can be continued. It is triggered by passing in the task order number or shelf number/AMR current location/equipment number. The priority of continuing the task is "task order number>equipment number>equipment current location>shelf number". When both the task order number and the equipment number are passed in, the RCS system triggers the task based on the task found by the task order number. When the RCS system has multiple areas, and there is a same point number in multiple areas, when passing in the current location of the equipment, that is, the point number, it is necessary to add areaId to distinguish the continued task.
- Usage scenario: The task execution is paused at a certain step. After the third-party operation is completed, let the AMR continue the task. It can also be used as a safety protection before entering the storage location. When the MES and other systems receive a task status of 10 waiting for confirmation, after confirming that it is safe to enter the storage location, they can call this interface to let the AMR enter the storage location.

2.1.2.2 Parameter details

Table 2-4 Details of external interface parameters for continuing tasks

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	orderId	string	100	No	●Trigger tasks through third-party task order numbers ● When performing wave tasks, the task must be triggered by subOrderId
	subOrderId	string	100	No	Subtask number: ● The business process template contains multiple RCS tasks, corresponding to the subOrderId of one of the RCS tasks ● For wave tasks, the task ID of a single business process in the wave; corresponding to the subOrderId of a single business when

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					the wave task is issued
	shelfNumber	String	32	No	Trigger mission by vehicle number ● Latent lifting AMR task, assigning shelf number ● For other AMR type tasks, the assignment is invalid.
	deviceNum	String	32	No	Device number: trigger the task by device number
	currentStation	String	32	No	Current point: trigger the task through the current point The QR code coordinates of the current point. If the point number exists in multiple areas, areaId needs to be passed in.
	areaId	int	10	No	Area ID, which needs to be passed in when there are multiple areas in the RCS system and there are the same point numbers in multiple areas.
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

2.1.2.3 Examples

ask:

```
{
"orderId":"WenDingXingTaskTest514"
}
```

Answer:

```
{
```

```
"code":1000,
"desc":"success"
}
```


2.1.3 Cancel task*

2.1.3.1 Interface details

- Interface Name:http://IP:7000/ics/out/task/cancelTask
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is called when the third-party system needs to actively cancel the task. If the task is unloaded when it is canceled, the AMR will be released directly. If the AMR is loaded, the processing after canceling the task currently supports the following three processing methods:
  - ◇ After canceling the task, AMR waits for on-site intervention. (Default mode)
  - ◇ After canceling the task, the AMR will move the shelf or pallet back to the starting point. (Configuration required)
  - ◇ Specify the target location when canceling the task, and AMR will move the goods to the specified location. (Configuration required)

  For detailed configuration, please refer to the RCS platform "Configuration Management > Task Template Configuration > RCS Task Template > Click the template number" to configure the load cancellation task strategy.
- Usage scenario: The upper-level system needs to cancel the task that has been sent to RCS.

2.1.3.2 Parameter details

Table 2-5 Cancel task interface parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	orderId	string	100	No	Third-party system task order number ● Supports batch cancellation of tasks. The format of the incoming data is an array. Please refer to the example below for details. ● Passing this value into a wave task means canceling all tasks under the wave.
	subOrderId	string	100	No	Subtask number: The business process template contains multiple RCS tasks,

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					corresponding to the subOrderId of one of the RCS tasks For wave tasks, the task ID of a single business process in the wave; corresponding to the subOrderId of a single business when the wave task is issued.
	deviceNumber	string	32	No	Equipment number, select one of the third-party system task order number. Only supports cancellation of third-party system task orders.
	destPosition	string	64	No	The target location where the shelf will be placed after the task is canceled ● The task has been executed until the shelf is lifted and the value passed in is valid, otherwise it will be canceled directly. ● If this value is not passed, the shelf will be processed according to the cancellation policy configured by the system by default.
Answer	code	int	6	Yes	Return status code: 1000: The operation is successful, and the cancellation result will be returned through the task status report. Other: Failed
	desc	string	64	Yes	Status code description

2.1.3.3 Examples

ask:

● Cancel based on device number

```
[
    {
        "deviceNumber": "wewf",
        "destPosition": "01001047"
    },
{
        "deviceNumber": "wewg",
        "destPosition": "01001048"
    }

]
```

● Cancel according to the task order number

```
[{
"orderId":"WenDingXingTaskTest514",
"destPosition":"Point 1"
},
{
"orderId":"WenDingXingTaskTest512",
"destPosition":"Point 2"
}]
```

说明

Batch task cancellation is supported. destPosition needs to be passed in only when specifying the target point to which the shelf will be moved after the task is canceled. It does not need to be passed in in other cases.

Answer:

```
 {
"code":1000,
"desc":"success"
}
```

2.2 Task status reporting interface

2.2.1 Interface details

- Interface name: provided by the third-party system (after the interface is implemented, please provide the interface name/interface address to the on-site implementation technicians in a timely manner, which needs to be configured to the ICS system)
- Provider: third-party system
- Caller: ICS integrated control system
- Interface protocol: HTTP
- Function description: The integrated control system proactively reports the task execution status to the third-party system.
- Usage scenario: The third-party system needs to receive the task status reported by the AMR and call it when performing related processing.
- Calling logic: The third-party system provides an interface URL, which is implemented according to the following protocol. RCS configures the task status to be reported to the third-party system at the node where the task status needs to be reported to the third-party system. When the AMR runs to the corresponding node, it will report the task status to the third-party system.
- Retry mechanism: If the third party does not respond with 1000, the RCS will retry according to the configured number of retries after receiving the request until the number is exceeded or the third party returns 1000 success.

2.2.2 Parameter details

Table 2-6 Task status interface parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	orderId	string	100	Yes	Third-party system task order number
	subOrderId	string	100	No	Subtask number ● The process template contains multiple subtasks. If you want to monitor the execution progress of each subtask, you can use this field. ● In the wave task, this field is used for each task number.

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
	modelProcessCode	string	100	Yes	Business process template number
	qrContent	string	100	No	The target point of the current subtask corresponds to the point issued when the task is issued.
	deviceNum	string	32	No	AMR Number
	deviceCode	string	64	No	AMR serial number
	status	int	2	Yes	Task status: 3: Cancelled 5: Send failed 6: Running 7: Execution failed 8: Completed 9: Issued 10: Wait for confirmation 20: Picking up goods 21: Pickup completed 22: Releasing goods 23: Release of goods completed Supports custom status, such as 100 means requesting the elevator to release the goods, 110 means the goods are successfully placed in the elevator, etc.
	storageNum	string	32	No	High-level warehouse number, corresponding to high-level warehouse location
	errorDesc	string	64	No	The reason why the task failed to execute or failed when sent to AMR
	shelfCurrPosition	string	32	No	Shelf current location
	shelfNumber	string	32	No	Vehicle No. ● When moving shelves, correspond to the

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					shelf numbers ● During CTU task, the number identified by the corresponding material box
	hasContainer	string	2	No	During CTU task inventory, whether the container is scanned: 0: No 1: Yes
	subTaskTypeId	String	6	Yes	RCSSubtask type, extended field, see"appendix2 Subtask type comparison table"
	subTaskSeq	String	2	Yes	RCS subtask number, starting from 0, the first subtask number is 0
	subTaskStatus	String	2	Yes	RCS subtask status: 1: Not started 2: Running* 3: In progress* 4: Failure 5: Cancel
	extraInfo	Json	400	No	The additional fields are mainly used for data transmission by third-party systems, and ICS does not perform any logical processing. For example: The task issued by the third-party system contains a material code, and the third-party system hopes that the material code will be included when the task is reported.
Answer	code	int	6	Yes	Return status code: 1000: Success

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					Other: Failed
	desc	string	64	Yes	Status code description

2.2.3 Examples

ask:


```
{
"storageNum": "C13",
"subTaskStatus": "3",
"orderId": "1637846849",
"deviceCode": "7J08197PA400001 ",
"modelProcessCode ": "F3SensorTriggerTaskProcess ",
"subTaskTypeId": "20",
"subTaskId": "2553259",
"deviceNum": "dn19",
"qrContent": "33000513",
"subTaskSeq": "0",
"status": 5,
"errorDesc": "33000513 The mission point does not exist",
      "shelfCurrPosition": "33000511",
      "extraInfo":{"sfgds":"321dfsa","dfs":"fdsg"}
}
```


Successful response:


```
 {
"code":1000,
"desc":"success"
}
```

Chapter 3 Optional Interfaces

3.1 Third-party calling interface provided by ICS system

3.1.1 Task related interfaces

3.1.1.1 Query task status

3.1.1.1.1 Interface details

- Interface Name:http://IP:7000/ics/out/task/getTaskOrderStatus
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used by the third-party system to actively obtain the AMR task status. All tasks that have been issued can obtain the current execution status of the task through this interface.
- Usage scenario: When ICS fails to report the task status for a certain period of time, the third-party system can actively query and obtain it.
- Call frequency: The interface call shall not exceed 50 times/second, and a single task number shall not exceed 10 seconds/time.

3.1.1.1.2 Parameter details

Table 3-1 Query task status interface parameter details

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
Request	orderId		string	100	Yes	The task order number specified when the third-party system issues the task
Answer	code		int	6	Yes	Return status code: 1000: Success Other: Failed
	data		JsonObject	1000	Yes	Returned task status information
	data	areaId	int	10	Yes	Region ID
		createTime	long	12	Yes	Task creation time:

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						timestamp
		fromSystem	string	50	Yes	Source system: The third-party system that issues tasks. Customized by the third-party system
		status	int	2	Yes	Business Task Force Status: 1: Not sent 3: Cancelled 4: Sending 5: Send failed 6: Running 7: Execution failed 8: Completed 9: Issued 10: Wait for confirmation Custom status is not supported
		taskOrderDetail	JsonArray	1000	Yes	Task status details
	taskOrderDetail	qrContent	string	100	No	Current target point of the mission
		deviceNum	string	32	No	AMR number for the task being performed
		deviceCode	String	32	No	AMR serial number that performs the task
		subOrderId	String	32	No	Sub-order number
		status	int	2	Yes	RCS task status: 1: Not sent 3: Cancelled 4: Sending 5: Send failed 6: Running

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						7: Execution failed 8: Completed 9: Issued 10: Wait for confirmation Support custom status
		errorDesc	string	64	No	Task execution failed/cancelled, or the reason for failure/cancellation when sent to AMR
		subTaskStatus	String	2	Yes	RCS subtask status 1: Not started 2: Running* 3: Completed 4: Failure 5: Cancel Custom status is not supported
		subTaskTypeId	String	6	Yes	RCSSubtask ClassType, extended fields, see"appendix2 Subtask type comparison table"
		subTaskId	String	10	Yes	RCS subtask number, alternate field
		subTaskSeq	String	2	Yes	RCS subtask sequence number, starting from 0, extended field
		time	long	13	Yes	System time of current status
		extraInfo	Json	400	No	Additional fields
	desc		string	64	Yes	Status code description

3.1.1.1.3 Examples

ask:

```
{
        "orderId":"WMS1100223"
 }
```

Answer:

```
{
"code": 1000,
"data": {
"areaId": 1,
          "createTime": 1684920666,
"taskOrderDetail": [
              {
                    "deviceNum": "ffff",
                    "qrContent": "0040",
                    "subTaskStatus": 3,
                    "subTaskSeq": 3,
                    "time": 1684920709,
                    "deviceCode": "ffff0000",
                    "subTaskTypeId": 5,
                    "subTaskId": 4,
                    "status": "8",
"extraInfo":{"sfgds":"321dfsa","dfs":"fdsg"}
              }
        ],
          "fromSystem": "call_task",
"status": 8
    },
"desc": "Request successful"
}
```

3.1.1.2 Change mission location

3.1.1.2.1 Interface details

- Interface Name:http://IP:7000/ics/out/task/updateOrderPointInfo
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used after the task is issued and the AMR has not yet reached the target point. At this time, the starting target point of the task needs to be changed, such as changing the transport task from point AB to point CD.
- Usage scenario: The task has been issued, but due to changes in business needs, the system needs to change the target point.

3.1.1.2.2 Parameter details

Table 3-2 Details of external interface parameters for continuing tasks

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	orderId	string	100	Yes	The task number specified when the third party issues the task
	pointPath	string	255	Yes	Task path points, the changed starting point and target point must be sent (multiple points are separated by English commas)
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.1.1.2.3 Examples

ask

```
 {
"orderId":"WenDingXingTaskTest514",
"pointPath":"80000231,80000212"
}
```

说明

- There are certain restrictions on changing the task point. For example, A→B. When the AMR reaches point A, the task can no longer be changed to C→D. However, the task can be changed to A→D. When the AMR reaches point B, the task point change will no longer be supported.
- Simply put, after a task is issued, when the AMR reaches a certain location, that location will no longer support changing the task point.
- When changing the mission point, the starting and target points required for the entire path need to be passed down.

Successful response:

```
 {
"code":1000,
"desc":"success"
}
```

There is no response to the order information:

```
{
"code":8007,
"data":"",
"desc":"The current order does not exist"
```

```
}
```

3.1.1.3 Update task priority

3.1.1.3.1 Interface details

- Interface Name:http://IP:7000/ics/out/updateTaskPriority
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: After a task is issued, if an AMR has not been assigned to execute a task, the order of task execution can be manually controlled to request this interface so that the task is assigned to AMR for execution first; it also supports lowering task priority.
- Usage scenario: When there are many tasks or urgent tasks, you want to adjust the priority of certain tasks and use it in priority execution scenarios. It is effective to call this interface before the task is assigned (task status is not started, issued) and AMR.

3.1.1.3.2 Parameter details

Table 3-3 Update task priority interface parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	orderId	string	100	Yes	orderId when the third-party system issues the task
	score	string	10	Yes	Range: between 500-100000. A single increase of 100000 will generally be executed immediately. The task allocation score is calculated based on a certain weight according to the task priority, task start and end points, and task issuance time. The larger the score, the higher the priority.
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.1.1.3.3 Examples

ask:

```
 {
"orderId":"chengpinchukufenjian01_20",
"score":"100000"
}
```

说明

It is useful to call this interface before AGV when the task has not been assigned (the task status is not started, but has been issued).

Answer

```
 {
"code":1000,
"desc":"success"
}
```

3.1.1.4 End the task

3.1.1.4.1 Interface details

- Interface Name:http://IP:7000/ics/out/endTask
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used by the third-party system to end the task ahead of schedule.
- Usage scenario: For scenarios where a third party wants to end the task early.

3.1.1.4.2 Parameter details

Table 3-4 Details of external interface parameters for ending tasks

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	orderId	string	100	Yes	Task order number: If you need to complete a task, just pass the task order number of the task.
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	data	string		Yes	Return result content
	desc	string	64	Yes	Status code description

3.1.1.4.3 Examples

ask:

```
 {
"orderId":"3215435"
}
```

Answer:

```
{
```

```
"code": 1000,
"desc": "Request successful"
}
```


3.1.1.5 Pre-scheduling tasks

3.1.1.5.1 Interface details

- Interface Name:http://IP:7000/ics/taskOrder/preAllocTask
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used when the customer system needs to issue tasks in advance, schedule AMR to the task point and wait for the task point to generate tasks, and then quickly respond to transportation.
  For example, if workbench A has a trailer of goods that needs to be unloaded in 1 minute, this interface can be used to send the task to point A, and the AMR will arrive at point A and wait within 1 minute.

说明

It is generally used in scenarios without identification preparation points, such as scenarios where latent lifting AMRs do not need shelf identification for transportation and scenarios where roller trucks call for help in advance. Scenarios that require identification preparation points are not currently supported, such as forklifts and latent lifting shelf identification scenarios. Box-type AMRs are not supported.

3.1.1.5.2 Parameter details

Table 3-5 Pre-scheduling task parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	webCode	string	10	Yes	Machine number
	taskCountDown	int	4	Yes	How long is it expected that a task will be generated at this point, in seconds?
	taskTyp	string	10	No	Task process template number ● It is recommended to specify and send this field; if the task process template number is not specified, the system default configuration

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					template is used. The system default configuration is a template named "move" which is consistent throughout the system and requires manual RCS configuration. ● If the task has requirements for the execution of AMR type or has a specified AMR number, this attribute must be issued.
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description
	data	Json	64	No	

3.1.1.5.3 Examples

ask:

```
 [
{
"webCode": "07170119",
"taskCountDown":100,
"taskTyp":"move"
}
]
```

Answer:

```
 {
"code": 1000,
"desc": "Request successful"
}
```

3.1.1.6 Rotating shelves

3.1.1.6.1 Interface details

- Interface Name:http://IP:7000/ics/out/changeShelfSide
  - ◇ In a scenario where picking is required, when the AMR is carrying the shelf and stops at the workbench waiting to continue, humans are picking goods. After picking one side, when the shelf needs to be rotated to pick the goods on the other side, request the RCS to rotate the AMR 180 degrees.

  ◇ Side A and B need to be configured by the delivery colleagues in the RCS background shelf_angle_config.
- Interface protocol: HTTP
- Provider: ICS integrated control system
- Caller: third-party system
- Usage scenario: Usually used in warehousing and sorting goods to facilitate workers to pick.

3.1.1.6.2 Parameter details

Table 3-6 Rotating shelf interface parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	orderId	string	100	No	The task order number when the task is issued by the third-party system
	subOrderId	String	100	No	The subtask number specified when the third-party system issues the task
	shelfNumber	String	12	No	Shelf number. You can choose either one of them or orderId. When both are entered, orderId will prevail.
	shelfToSide	string	64	Yes	Shelf orientation after rotating shelf ● A/B side can be configured, and the angle can also be directly issued ● The shelf angle of the A/B surface corresponding to this workbench needs to be configured on the RCS A: Side A faces the

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					workers' working position B: Side B faces the workers' working position
	toLocationCode	string	64	Yes	The picking table to go to after rotation
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.1.1.6.3 Examples

ask:

```
 {
"orderId":"chengpinchukufenjian01_20",
"shelfToSide":"A",
"toLocationCode":"81001945"
}
```

Answer:

```
 {
"code":1000,
"desc":"success"
}
```

3.1.1.7 Query task list

3.1.1.7.1 Interface details

- Interface Name:http://IP:7000/ics/out/task/getOrderList
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: Query the task information currently being executed in the specified area.
- Usage scenario: Used when you need to obtain specified or all task information.
- Calling frequency: It is recommended not to exceed 20 times/second

3.1.1.7.2 Parameter details

Table 3-7 Query task list interface parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	areaId	int	10	Yes	Region id
	templateCode	String	12	No	RCS task template number, multiple selections allowed, separated by commas
	OrderStatus	String	64	No	Task status: 1: Not sent 2: Canceling 3: Cancelled 4: Sending 5: Send failed 6: Executing 7: Execution failed 8: Mission accomplished 9: Issued Supports multiple selections, separated by commas; does not support custom status
	Priority	String	64	No	Priority: 4: High 6: Medium 8: Low Support multiple selection, comma separated
	qrName	String	64	No	Node name/point QR code, multiple selections are not

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					supported
	shelfNum	String	64	No	Shelf number, multiple selections are not supported
	storageNum	String	64	No	Storage location number, multiple selections are not supported
	orderIdList	JsonArray	64	No	Order id list, order id is String, separated by commas
	taskIdList	JsonArray	64	No	Task id list, task id is String, separated by commas
	intendStartTimeHigh	String	64	No	Upper limit of screening time, time format: YY-MM-DD HH-MM-SS
	intendStartTimeLow	String	64	No	Filtering time lower limit, time format: YY-MM-DD HH-MM-SS
	StartPage	String	64	No	Starting Page
	pageSize	String	64	No	Number of entries per page
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed

Parameter type	Parameter name			Field Type	Maximum length	Is it required?	Description
	desc			String	64	Yes	Status code description
	data			JsonObject	64	Yes	Return information
	data	Tasks		JsonArray	64	Yes	Task list array
		Tasks	TemplateCode	String	64	Yes	Task template number
			TemplateName	String	64	Yes	Task template name
			RobotType	String	64	Yes	Device type
			AgvNum	String	32	No	Equipment number
			TaskId	int	10	Yes	RCS task id
			ZoneId	int	10	Yes	Sub-area id
			Priority	int	2	Yes	Priority: 4: Low 6: Medium 8: High
			OrderPath	String	64	Yes	Task Path
			OrderId	String	100	No	The order number orderId of the task issued by the third-party system. If the task is not issued by a third party, this field will be empty.
			Materiel	String	64	No	Material information
			OrderStatus	int	64	Yes	Order Status
			createTime	String	64	Yes	Creation time
			ActualStartTime	String	64	No	Task start time

Parameter type	Parameter name			Field Type	Maximum length	Is it required?	Description
			ActualEndTime	String	64	No	Task end time
			TaskTime	String	64	Yes	Execution time, in seconds
			ErrorCode	String	6	No	Task failure error code, has value when failed
			ErrorDesc	String	64	No	Task failure error code description, has value when failed
		TotalPage		int	4	Yes	Total pages
		pageSize		int	4	Yes	Maximum number of entries per page
		Page		int	4	Yes	Current Page Number
		totalCount		int	4	Yes	Total number of items

3.1.1.7.3 Examples

ask:

说明

Query the entire task list without adding any filtering conditions.

```
 {
"areaId": 1
}
```

Add filter conditions to filter eligible tasks:

```
{
"areaId": 1,
        "templateCode": "templateCode1",
"OrderStatus": "6",
"Priority": "6",
"qrName": "00000183",
"shelfNum": "shelf0",
"storageNum": "H001",
        "orderIdList": ["103","104"],
```

```
        "taskIdList": ["2101","2102"],
"batchIdList": [],
        "intendStartTimeHigh": "2023-05-24 23:59:59",
        "intendStartTimeLow": "2023-05-23 00:00:00",
"StartPage": "1",
"pageSize": "20"
}
```


Answer:


```
{
"code": 1000,
"data": {
"Tasks": [
        {
                "AllowChangeDevice": 1,
                "RobotType": "RTA-C350-L-4L4T-D",
                "TaskId": 7944,
                "OrderStatus": 8,
"ErrorDesc": "ICS-BUSINESS requested to end task",
"TemplateName": "Coiling and unloading",
                "Priority": 6,
                "OrderPath": "10000037—10000037—11000143",
                "OrderId": "1_2_CT4_20241129190054526FZK",
                "Materiel": "[]",
                "TemplateCode": "DDownFull",
                "ActualStartTime": "2024-11-29 19:00:55",
                "FixedRobotId": "C3500001",
                "createTime": "2024-11-29 19:00:54",
                "ActualEndTime": "2024-11-29 19:11:14",
                "subType": 0,
                "TaskTime": "619",
                "AgvNum": "3501"
        }
    ],
"TotalPage": 1,
"pageSize": 20,
"Page": 1,
"totalCount": 1
  },
"desc": "Request successful"
}
```


3.1.2 Device related interfaces

3.1.2.1 Get AMR status

3.1.2.1.1 Interface details

- Interface Name:http://IP:7000/ics/out/device/list/deviceInfo
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP

- Function description: Obtain AMR operation status information, such as AMR location, equipment number, equipment serial number, equipment status and other information.
- Usage scenario: The third-party system needs to know the detailed operation information of the AMR equipment, such as the use of data dashboard.
- Calling frequency: It is recommended not to exceed 10 times/second. When used for real-time location display of AMR, the front-end page moves smoothly.

3.1.2.1.2 Parameter details

Table 3-8 Get AGV status interface parameter details

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
Request	areaId		int	10	No	Area Id: Provided by the integrated control system, this field is mainly used to distinguish different map areas. Default value: default.area.id: 1
	deviceType		int	2	Yes	Fixed value: 0
	deviceCode		string	64	No	Device serial number, supports fuzzy query. Do not query all
Answer	code		int	6	Yes	Return status code: 1000: Success Other: Failed
	desc		string	64	Yes	Status code description
	data		JsonArray		Yes	Returned device type information
	data	deviceCode	string	32	Yes	Device serial number
		deviceName	string	32	Yes	Equipment number
		odom	float	10	No	Equipment mileage, unit: M
		payLoad	string	4	No	Equipment load

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						status
		orderId	string	100	No	The task order number currently being executed (issued by a third-party system)
		taskPath	string	200	No	Task path point set, multiple points are separated by English commas
		shelfNumber	string	64	No	The carrier number being transported is the shelf number being transported by the potential lifting AMR, or the box number being transported by the box AMR.
		devicePositionRec	Array	32	No	The x,y coordinates of the device's real-time location. The first value is x and the second value is y.
		devicePosition	string	32	Yes	The (QR code) node number where the device is located. If it is not on the (QR code) node, the previous (QR code) node number will be reported.
		battery	string	4	Yes	Battery level
		state	string	32	Yes	Device status: Idle: Idle Initializin: Initializing

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						InTask: In Task Fault: Fault Offline: Offline InCharging: Charging InUpgrading: Upgrading
		oritation	Int	6	No	Equipment angle, unit: thousandths
		speed	int	6	No	Speed, unit: mm/s

3.1.2.1.3 Examples

ask:

```
 {
"areaId":"1",
"deviceType":"0"
}
```

Successful response:

```
{
"code": 1000,
"data": [
        {
                "taskPath": "55300593,03200340,55300593",
                "payLoad": "0.0",
                "orderId": "DWCS00160521",
                "shelfNum": "AG02-062",
                "devicePosition": "55300593",
                "devicePostionRec": [
                    136771,
                    54039
            ],
                "state": "InTask",
                "deviceCode": "6L0F5E5PA400002",
"battery": "59",
                "deviceName": "0001",
                "odom": "1001.349"
        }
    ],
"desc": "success"
}
```

3.1.2.2 Obtain real-time equipment alarm and detailed planning information

3.1.2.2.1 Interface details

- Interface Name:http://IP:PORT/getAGVTaskInfo
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is called when a third-party system wants to obtain the planning path information of a certain device.
- Usage scenario: used when the third-party system needs to obtain real-time alarm information and detailed planning information of AMR equipment.

3.1.2.2.2 Parameter details

Table 3-9 Interface parameter details for obtaining device real-time alarm and detailed planning information

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
Request	deviceCode		string	64	No	Device serial number
	deviceNumber		string	32	No	Device number, at least one of the device number and device serial number must be passed.
Answer	code		int	6	Yes	Return status code: 1000: Success Other: Failed
	desc		string	64	Yes	Status code description
	data		JsonArray		Yes	Returned device type information
	data	alarmMsg	string	32	No	Equipment alarm information
		data	string	4	No	Material information, reserved fields
		targetPositionName	string	32	No	The target point of the RCS mission currently

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						executed by this device
		advice	string	32	No	Current processing recommendations
		remark	string	4	No	Device current path status, algorithm status, device status, and planning status
		startPositionName	string	32	No	The target point of the RCS mission currently executed by this device
		taskTypeName	string	2	No	RCS subtask type name
		startPosition	string	32	No	QR code of the starting point of the current path segment
		targetPosition	string	32	No	QR code of target point of current path
		outOrderId	string	100	No	The task order number when the third-party system issues the task will have a value when the task is executed.
		taskTemplateName	string	128	No	RCS task template name
		state	string	2	Yes	Device status: 0: Idle 1: Moving 2: Executing action 3: Wait for confirmation 4: Load abnormality 5. Path

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						abnormality 6: Storage location abnormality 7: Charging abnormality 8: Failure 9: Offline 10: Charging 11: Upgrading 12: Download the upgrade package 13: Under maintenance 14: Initializing 15: Unknown status
		taskId	string	32	No	RCS task order number

3.1.2.2.3 Examples

ask:

```
{
"deviceCode":"7B03A73PA400010"
}
```

Answer:

```
{
"code": 1000,
"data": {
"alarmMsg": "Task 24@80937@461103 is not completed",
            "data": "[{\"materielNum\":\"\",\"materielName\":\"\",\"materielAccount\":\"\"}]",
            "targetPositionName": "",
"advice": "Check if the task is completed",
"remark": "[Equipment segment path status: algo24_442_869 segment path is being executed;
Algorithm status: in task, running normally; Equipment status: normal; Planning status:
plannable status; ]",
            "startPositionName": "",
"taskTypeName": "Mobile shelves/pallets (load arrival)",
            "startPosition": "77000005",
"alarmFlag": 1,
            "targetPosition": "77000005",
"outOrderId": "",
"taskTemplateName": "QR code outbound sorting",
```

```
            "shelfNumber": "99000102",
"state": 1,
"taskId": "80938"
      },
"desc": "success"
}
```


3.1.2.3 Get device task path

3.1.2.3.1 Interface details

- Interface Name:http://IP:7000/ics/out/getRobotTaskPath
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used to obtain the details of the specified AMR task path in the customer system.

3.1.2.3.2 Parameter details

Table 3-10 Get device task path interface parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	areaId	int	10	No	Area Id: Provided by the integrated control system, this field is mainly used to distinguish different map areas. Default value: default.area.id:1
	deviceCode	string	64	No	Device serial number, the device whose path needs to be obtained, multiple devices are supported, separated by English commas
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description
	areaInfo	jsonArray	64	Yes	Area information details
	areaId	string	64	Yes	Region id
	devices	jsonArray	64	Yes	Device information

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
	deviceNum	string	32	Yes	Equipment number
	pathNodes	jsonArray	1000	Yes	Path details, the four values represent the QR code value, x coordinate, y coordinate, and angle respectively.
	pathLockRects	jsonArray	1000	Yes	Path safety runs within a rectangular lock grid area, a set of 4 x/y coordinates
	pathLockCircs	jsonArray	1000	Yes	The circular lock area for safe operation of the equipment, x/y/r represents the coordinates of the center of the circle and the radius of the circle
	deviceCode	string	64	Yes	Device serial number

3.1.2.3.3 Examples

ask:

```
{
"areaId": 10,
      "deviceCode": "mmC0600000"

}
```

Answer:

```
{
"code": 1000,
"data": {
"areaInfo": [
{ "areaId": 2,
"devices": [
                    {
                              "deviceNum": "1000",
                              "pathNodes": [
                               "11000020,2470,10350,999000",
                               "11000032,3820,10350,999000",
                               "11000031,3820,9000,999000",
                               "11000007,1120,9000,999000"
                         ],
                              "pathLockRects": [
                               "4135,9855,3505,9855,3505,10845,4135,10845"
```

```
                                ],
                                    "pathLockCircs": [],
                                    "deviceCode": "mmC0600000"
                            }
                        ]
                    }
                ]
        },
"desc": "Request successful"
}
```


3.1.2.4 Control device pause/resume

3.1.2.4.1 Interface details

- Interface Name:http://IP:7000/ics/out/controlDevice
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used by the customer system to control AMR to stop or resume operation.

3.1.2.4.2 Parameter details

Table 3-11 Control device pause/resume interface parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	areaId	int	10	No	Area Id: Provided by the integrated control system, this field is mainly used to distinguish different map areas. Default value: default.area.id:1
	deviceNumber	string	32	No	Device number, multiple device numbers are separated by English commas. If it is empty, all devices in the area will be paused or resumed.
	orderId	string	100	No	Order number, supports pausing the execution of the ordered vehicle according to the order number to achieve the task suspension.
	all	int	2	Yes	Whether to suspend or resume all devices in this area 0:None

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					1: all
	controlWay	int	2	Yes	0: Pause the device 1: Restore the device
	stopType	int	2	No	Device pause mode 0: Stop smoothly to the next QR code (default is 0 if not transmitted) 1: Stop immediately 2: Stop with deceleration configured by RCS
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	data	String		Yes	Return content results
	desc	string	64	Yes	Status code description

3.1.2.4.3 Example (pause some devices)

ask:

```
{
"areaId": 10,
   "deviceNumber":"device1,device2",
"orderId":"task1001",
"all":0,
"controlWay": 0
}
```

Answer:

```
{
"code": 1000,
"data": ""
"desc": "Request successful"
}
```

3.1.2.4.4 Example (pause all devices)

ask:

```
{
"areaId": 10,
"deviceNumber":"",
"all":1,
"controlWay": 0
```

```
}
```

Answer:

```
{
"code": 1000,
"data": "",
"desc": "Request successful"
}
```

3.1.2.5 Release device pre-occupancy

3.1.2.5.1 Interface details

- Interface Name:http://IP:7000/ics/out/removeAGV
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used to release the device from pre-occupancy by the third-party system.
- Usage scenario: For scenarios where a third party wants to actively release a device. When a third party specifies that the device will not be released after the task is completed, and for some reason the device is no longer needed to continue executing the task, the device needs to be actively released.

3.1.2.5.2 Parameter details

Table 3-12 Details of parameters for releasing device pre-occupancy of external interfaces

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	deviceNumber	string	32	Yes	Device number: Use English commas to separate multiple devices.
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	data	string		Yes	Return result content
	desc	string	64	Yes	Status code description

3.1.2.5.3 Examples

ask:

```
{
"deviceNumber":"1112,2223"
```

```
}
```

Answer:

```
{
"code": 1000,
"data": "",
"desc": "Request successful"
}
```

3.1.2.6 Send charging

3.1.2.6.1 Interface details

- Interface Name:http://IP:7000/ics/out/gocharging
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used to send AMR charging tasks.

3.1.2.6.2 Parameter details

Table 3-13 Parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	deviceNumber	string	32	Yes	Equipment number
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.1.2.6.3 Examples

ask:

```
{
"deviceNumber": "1122"
}
```

Answer:

```
{
"code": 1000,
"desc": "Request successful"
}
```

3.1.2.7 Stop charging

3.1.2.7.1 Interface details

- Interface Name:http://IP:7000/ics/out/finishCharging
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used to end charging of AMR.

3.1.2.7.2 Parameter details

Table 3-14 Stop charging interface parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	deviceNumber	string	32	Yes	Equipment number
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.1.2.7.3 Examples

ask:

```
{
"deviceNumber": "1122"
}
```

Answer:

```
{
"code": 1000,
"desc": "Request successful"
}
```

3.1.3 Storage location related interfaces

3.1.3.1 Get shelf location

3.1.3.1.1 Interface details

- Interface Name:http://IP:7000/ics/out/shelf/getShelfPosition

- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used by the third-party system to obtain the shelf location maintained by RCS.
- Usage scenario: The shelf location is maintained by RCS, and the third-party system needs to synchronously obtain all shelf location information.
- Calling frequency: It is recommended not to exceed 20 times/second

3.1.3.1.2 Parameter details

Table 3-15 Get shelf location interface parameter details

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
Request	areaId		int	10	No	Area Id: Provided by the integrated control system, this field is mainly used to distinguish different map areas. Default value: default.area.id: 1
	shelfNums		string	100	No	Shelf number collection, multiple shelf numbers, separated by English commas
Answer	code		int	6	Yes	Return status code: 1000: Success Other: Failed
	desc		string	64	Yes	Status code description
	data		jsonArray		Yes	Get the shelf location information; if the shelf does not exist, data returns empty.
	data	shelfNum	string	10	Yes	Shelf Number
		shelfCurrStation	string	10	Yes	The QR code location of the shelf. If the shelf has no

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						location in the RCS system, return -1.
		posX	int	10	No	The x coordinate of the shelf. If the shelf has no position in the RCS system, return -1000000
		posY	Int	10	No	Shelf y coordinate. If the shelf has no position in the RCS system, return -1000000.
		shelfType	Int	6	No	Shelf Type
		angle	string	10	No	Shelf angle, if the shelf has no position in the RCS system, returns -1000

3.1.3.1.3 Examples

ask:

```
{
"areaId":"3",
"shelfNums":"shelf01,test"
}
```

Answer:

```
{
"code": 1000,
"data": [
        {
                "angle": "-1000",
                "posX": -1000000,
                "posY": -1000000,
                "shelfCurrStation": "-1",
                "shelfNum": "shelf01",
"shelfType": 1
        },
        {
"angle": "0",
"posX": 12160,
```

```
"posY": 14400,
                    "shelfCurrStation": "01000015",
                    "shelfNum": "test",
"shelfType": 1
            }
      ],
"desc": "Request successful"
}
```


3.1.3.2 Change shelf location

3.1.3.2.1 Interface details

- Interface Name:http://IP:7000/ics/out/shelf/changePosition
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used to update the shelf location.
- Usage scenario: The shelf location is maintained by RCS. When a person or a third-party system changes the location of a shelf, the information is synchronized to RCS.

3.1.3.2.2 Parameter details

Table 3-16 Update shelf location parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	areaId	int	10	No	Area Id: Provided by the integrated control system, this field is mainly used to distinguish different map areas. Default value: default.area.id:1
	shelfNum	string	10	Yes	Shelf Number
	destPosition	string	10	Yes	The target location to bind to
	action	Int	2	Yes	1: Bind shelf 2: Unbind the shelf
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.1.3.2.3 Examples

ask:

```
[
        {
"areaId": 1,
                "destPosition": "66000079",
                "shelfNum": "s001",
"action": 1
        }
]
```

Answer:

```
{
"code": 1000,
"desc": "Request successful"
}
```

3.1.3.3 Get all storage location information

3.1.3.3.1 Interface details

- Interface Name:http://IP:7000/ics/out/getStockStatus
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: In some scenarios where RCS does not record shelf location information, RCS may record whether there is a shelf at a certain location based on the task execution status. RCS calls this storage location information. When a third-party system needs this information, it can obtain it through this interface.
- Usage scenario: RCS records the empty and full status of storage locations based on tasks, and is used when a third-party system needs to obtain the status of these storage locations.
- Calling frequency: It is recommended not to exceed 20 times/second

3.1.3.3.2 Parameter details

Table 3-17 Interface parameter details for obtaining storage location information

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	areaId	code	10	No	Area Id: Provided by the integrated control system, this field is mainly used to distinguish different map areas. Default value: default.area.id:1
	pageSize	string	6	No	The number of items per page, if not filled in, the default is 20 items per query

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
	pageNo	string	6	No	Which page should I start searching from? If left unfilled, the query will start from page 1 each time.
	areaName	string	12	No	The name of the slice area can be checked in the slice area configuration page of the RCS client.
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description
	data	JsonObject		Yes	Feedback content
	pageSize	Int	5	No	Number of pages
	pageNo	Int	5	No	Page No.
	totalPage	Int	5	No	Total pages
	totalNum	Int	10	No	Total number of items
	stockList	JSONArray			Storage location information collection
	areaId	Int	10	Yes	The area id to which this storage location belongs
	inTask	Int	2	Yes	Has the task been triggered at this point?
	qrContent	String	32	Yes	QR code corresponding to the storage location
	stockStatus	Int	2	Yes	Storage location status, usually 0 corresponds to an empty storage location, and 2 corresponds to a full storage location.

3.1.3.3.3 Examples

ask:

```
{
"areaId":"1",
"pageSize":"2",
"pageNo":"1",
"areaName":"FC05"
}
```

Answer:

```
{
"code": 1000,
"data": {
"stockList": [
                {
                        "areaId": "1",
                        "inTask": "0",
                        "qrContent": "80000005",
                        "stockStatus": 0
                }
            ],
"totalNum": 2,
"totalPage": 1,
"pageNo": 1,
"pageSize": 2
      },
"desc": "success"
}
```

3.1.3.4 Change storage location status

3.1.3.4.1 Interface details

- Interface Name:http://IP:7000/ics/stock/update/appStockStatus
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used to manually synchronize the storage location change information to RCS after the third-party system manually intervenes in the storage location status.
- Usage scenario: When RCS does not maintain the shelves, but RCS maintains the empty and full conditions of the storage location based on the task status. After the storage location status is manually changed, it is necessary to manually maintain and change the data and synchronize the changes to RCS.

3.1.3.4.2 Parameter details

Table 3-18 Update storage location status details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	qrContent	string	32	Yes	Storage location number ● Plane storage location QR code node number ● If the storage location is a high-position storage, then the storage location number corresponding to the high-position storage is
	nodeStatus	string	2	Yes	Storage location status: 0: Empty storage location 2: Full storage space
	extendInfo	Json	400	No	Additional fields ● Dense library: can be used to determine point selection based on additional attributes passed in, such as passed in material information ● Non-dense library: This attribute is only processed transparently)
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.1.3.4.3 Examples

ask:

```
{
"qrContent":"80003212",
"nodeStatus":"0"
}
```

Answer:

```
{
"code":1000,
"desc":"success"
}
```

3.1.4 Traffic control related interfaces

3.1.4.1 Application for entry and exit of traffic control area

3.1.4.1.1 Interface details

- Interface Name:http://IP:7000/traffic/control
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Functional description: This interface is the ICS integrated control system as the supervisory party of the traffic control zone, the third-party system as the managed party of the traffic control zone, and the third party applies to enter or leave the traffic control zone.

3.1.4.1.2 Parameter details

Table 3-19 Parameter details for applying to enter the traffic control area

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	reqCode	string	10	Yes	Apply for an order number. One request is for the same order number. This order number can be used for multiple requests during the process of multiple AMRs passing through.
	deviceNumber	string	32	Yes	AMR equipment number
	askZoneCode	string	10	Yes	Traffic control area number you are applying to enter
	action	int	1	Yes	1: Request entry

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					2: Apply to leave
Answer	code	int	6	Yes	Return status code: 1000: Success, run and enter Other: Failed, No Entry
	desc	string	64	Yes	Status code description
	data	Json	64	No	

3.1.4.1.3 Examples

ask:

```
{
"reqCode": "1235435653",
"deviceNumber":"1001",
"askZoneCode":"MAREA12",
"action": 1
}
```

Answer:

```
{
"code": 1000,
"desc": "Request successful"
}
```

3.1.4.2 Enable/disable no-go zones

3.1.4.2.1 Interface details

- Interface Name:http://IP:7000/ics/out/enableForbiddenZone
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is used when the customer system needs to control the enable/disable of the prohibited area.

3.1.4.2.2 Parameter details

Table 3-20 Parameter details for enabling or disabling restricted areas

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	indBind	int	1	Yes	Enable disable flag:

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					1: Enable no-go zone 0: Disable no-go zones
	matterArea	string	64	Yes	No driving zone name
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.1.4.2.3 Examples

ask:

```
{
"indBind": 1,
"matterArea":"No entry area 1"
}
```

Answer:

```
{
"code": 1000,
"desc": "Request successful"
}
```

3.1.5 Get map interface

3.1.5.1 Interface details

- Interface Name:http://IP:7000/ics/out/topologyList
- Provider: ICS integrated control system
- Caller: third-party system
- Interface protocol: HTTP
- Function description: This interface is called when the client system needs to obtain a map file and returns a map download address.

3.1.5.2 Parameter details

Table 3-21 Get map parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	areaId	int	10	No	Area id number, if not filled in, the default value is 1, which can be modified through ICS

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					configuration. The default value is: default.area.id:1
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description
	data	Json	64	No	Json object
	mapZipUrl	string	5000	No	Download address of the entire map package compressed file
	mapJsonUrl	string	5000	No	Map Json topology file download address
	mapId	int	10	No	Map number, same as area id

3.1.5.3 Examples

ask:

```
 {
"areaId": 8
}
```

Answer:

```
{
"code": 1000,
"data": {
            "mapZipUrl":
"http://10.55.216.75:8888/textureMap/8/default/Map/commonTopo.zip?token=3ndR1SMbDF
70DRzS3b0jSD8KafHVxrzE",
"mapId": 8,
            "mapJsonUrl":
"http://10.55.216.75:8888/textureMap/8/default/Map/compress.json?token=ZtEs6cWb54jVna
bmfZoC623bBDR9Vw9J"
      },
"desc": "Request successful"
}
```

3.2 Interfaces provided by third-party systems

3.2.1 Report device status to third-party system

3.2.1.1 Interface details

- Interface name: provided by the third-party system (after the interface is implemented, please provide the interface name/interface address to the on-site implementation technicians in a timely manner, which needs to be configured to the ICS system)
- Provider: third-party system
- Caller: ICS integrated control system
- Interface protocol: HTTP
- Usage scenario: When the integrated control system needs to actively report the AMR status to a third-party system, this interface of the third-party system is called.
- Calling logic: Polling reporting, reporting every 2 seconds by default, the interface can be configured and modified

3.2.1.2 Parameter details

Table 3-22 AMR equipment status parameter details

Request parameters	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	deviceCode	string	32	Yes	Device serial number
	payLoad	string	4	No	Equipment load status
	devicePositionRec	Array	32	No	The x,y coordinates of the QR code where the device is located
	devicePosition	string	32	Yes	Device current location
	battery	string	4	Yes	Battery level
	deviceName	string	32	Yes	Equipment number
	state	String	32	Yes	Idle: Idle Initializin: Initializing InTask: In Task Fault: Fault Offline: Offline InCharging: Charging InUpgrading: Upgrading
	oritation	int	6	No	Device angle, in thousandths

Request parameters	Parameter name	Field Type	Maximum length	Is it required?	Description
	orderId	string	100	No	Task order number when a third party issues a task
	shelfNumber	string	64	No	Vehicle No. ● The potential lifting AMR is the number of the shelf being transported. ● The material box AMR is used to transport material boxes
	speed	int	6	No	Speed, unit: mm/s
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.2.1.3 Examples

ask:

```
[
    {
"taskPath": "01000037,01000040",
"payLoad": "1.0",
"orderId": "5975858_test_15_test36",
"shelfNum": "TEST111",
"devicePosition": "01001045",
"devicePostionRec": [5410, 2250],
"shelfNumber": "TEST111",
"state": "InTask",
"deviceCode": "ffff0000",
"battery": "27",
"deviceName": "ffff",
}
]
```

Answer:

```
{
"code": 1000,
"desc": "Request successful"
```

```
}
```

3.2.2 Request a third-party system to obtain the task location during the task

3.2.2.1 Interface details

- Interface name: provided by the third-party system (after the interface is implemented, please provide the interface name/interface address to the on-site implementation technicians in a timely manner, which needs to be configured to the ICS system)
- Provider: third-party system
- Caller: ICS integrated control system
- Interface protocol: HTTP
- Functional description: When the integrated control system needs a task point during a task, it requests the third-party system, and the third-party system returns a task point or a piece of area.
- Usage scenario: Task A to B transportation, when issuing the task, only point A is issued. After the AMR reaches point A, it requests the third-party system to obtain the target point B. The third-party system returns the target point B to the ICS. If there is no target point currently, it can wait until there is a target point before returning. Before the third-party system returns the target point, the AMR will wait in place. If the third-party system determines that there will be no more target point and does not need to continue the task, the task cancellation interface can be called to end the task.
- Call frequency: When the third-party system does not return the target point, retry the application once every 2 seconds.

3.2.2.2 Parameter details

Table 3-23 Obtaining mission point parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	orderId	string	100	Yes	The task order number when the task is issued by the third-party system
	subOrderId	String	100	No	Third party system sub-order id
	modelProcessCode	string	100	Yes	Business process template number
	queueId	string	100	No	queue id For material box

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					AMR tasks, multiple tasks of the same material box AMR are represented by the same queue ID. The third-party system can use this to assign adjacent storage locations to these queue tasks to optimize efficiency.
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description
	data	JsonObject		Yes	Returned mission point information
	pointName	string	10	Yes	The returned mission point, if it returns to the area, the RCS will autonomously select a point that meets the conditions in the area.

3.2.2.3 Examples

ask:

```
{
"orderId":"task1111",
"modelProcessCode":"kongkuangfanh/ui"
}
```

Answer:

```
{
"code":1000,
```

```
"desc":"success",
"data":{
            "pointName":"pointA"
}
}
```


3.2.3 Alarm message reporting

3.2.3.1 Interface details

- Interface name: provided by the third-party system (after the interface is implemented, please provide the interface name/interface address to the on-site implementation technicians in a timely manner, which needs to be configured to the ICS system)
- Provider: third-party system
- Caller: ICS integrated control system
- Interface protocol: HTTP
- Usage scenario: When a third party needs to know the AMR alarm message of RCS, RCS generates an alarm message and actively reports it to the third-party system.
- Calling logic: Report when an alarm occurs, report once every 2s by default, and the interface can be configured and modified.

3.2.3.2 Parameter details

Table 3-24 Alarm information parameter details

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	deviceNum	string	32	No	Equipment number
	alarmDesc	string	500	Yes	Alarm description
	alarmType	int	6	Yes	Alarm type, see"appendix1 Equipment alarm type comparison table"
	areaId	int	10	Yes	Region id
	alarmReadFlag	int	1	Yes	Has it been read?
	channelDeviceId	string	32	No	Alarm location
	alarmSource	string	6	Yes	Alarm system source
	channelName	string	256	No	Recommended treatment
	alarmDate	string	15	Yes	Alarm date
	deviceName	string	15	No	Device serial number
	alarmGrade	int	2	Yes	Alarm level: 0,1,2. The higher the number, the more serious it is.

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
	alarmStatus	int	2	Yes	Alarm status 0: Alarm in progress 1: Alarm status restored
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.2.3.3 Examples

ask:

```
{
"deviceNum": "jbz1",
"alarmDesc": "Device offline",
"alarmType": 1,
"areaId": 13,
"alarmReadFlag": 1,
"channelDeviceId": "80000292",
"alarmSource": "rcs",
"channelName": "1. Check the network status \\r\\n 2. Restart the device",
"alarmDate": 538227704,
"deviceName": "jbztest0000",
"alarmGrade": 2
}
```

Answer:

```
{
"code": 1000,
"desc": "Request successful"
}
```

3.2.4 Actively report shelf location to the third-party system

3.2.4.1 Interface details

- Interface name: provided by the third-party system (after the interface is implemented, please provide the interface name/interface address to the on-site implementation technicians in a timely manner, which needs to be configured to the ICS system)
- Provider: Third-party system
- Caller: ICS integrated control system
- Interface protocol: HTTP

- Usage scenario: The shelf location is maintained and managed by RCS, and the third-party system needs the real-time shelf location of RCS
- Calling logic: Polling reporting, reporting every 2 seconds by default, the interface can be configured and modified

3.2.4.2 Parameter details

Table 3-25 Details of actively reporting shelf location parameters

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	shelfNum	string	10	Yes	Shelf Number
	shelfCurrStation	string	10	Yes	The QR code location of the shelf. If the shelf has no location in the RCS system, -1 is returned and there is no value for x/y below.
	posX	int	10	No	Shelf x coordinate
	posY	Int	10	No	Shelf y coordinate
	shelfType	Int	6	No	Shelf Type
	angle	string	10	No	Shelf Angle
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.2.4.3 Examples

ask:

```
[
    {
            "shelfNum":"66000454",
            "shelfCurrStation": "66000079",
"posX": 32434,
"posY":12313,
"shelfType":1,
"angle":0
  }
```

Answer:

```
{
"code": 1000,
```

"desc": "Request successful"
}

3.2.5 Actively report the charging pile status to the third-party system

3.2.5.1 Interface details

- Interface name: provided by the third-party system (after the interface is implemented, please provide the interface name/interface address to the on-site implementation technicians in a timely manner, which needs to be configured to the ICS system)
- Provider: Third-party system
- Caller: ICS integrated control system
- Interface protocol: HTTP
- Usage scenario: The third-party system needs to know the status details of the charging pile.
- Calling logic: Polling reporting, reporting every 2 seconds by default, the interface can be configured and modified.

3.2.5.2 Parameter details

Table 3-26 Actively report charging pile status

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	deviceCode	String	30	Yes	Charging station serial number
	deviceName	String	10	Yes	Charging station number
	state	String	20	Yes	Charging station status: Idle Offline InCharging Charging Fault Docked
	chargeMode	String	10	Yes	
	enable	Int	4	Yes	Is it enabled? 1 Enable 0 Disable
	chargeCoordinate	JsonArray		Yes	Coordinate 2D array, x, y
	chargePoint	String	10	Yes	Charging point QR code coordinates
	bindAMRNum	String	10	No	When the charging pile is docked for charging, the AMR device

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
					number of the charging pile is docked
	bindAMRCode	String	30	No	When the charging pile is docked for charging, the serial number of the AMR device docked with the charging pile
Answer	code	int	6	Yes	Return status code: 1000: Success Other: Failed
	desc	string	64	Yes	Status code description

3.2.5.3 Examples

ask:

```
[
    {
        "chargeCoordinate": [
            37598,
            3300
        ],
        "chargeMode": "Auto",
"enable": 1,
        "deviceCode": "2222CP-0000",
"state": "Idle",
        "deviceName": "ewew",
        "chargePoint": "10000061"
    },
    {
        "chargeCoordinate": [
            38800,
            3300
        ],
        "chargeMode": "Auto",
"enable": 1,
        "deviceCode": "2222CP-0001",
"state": "Idle",
        "deviceName": "gSxI",
        "chargePoint": "10000064"
    }
]
```

Answer:

```
{
"code": 1000,
"desc": "Request successful"
```

3.2.6 Actively report the scanning information of the scanning door to the third-party system

3.2.6.1 Interface details

- Interface name: provided by the third-party system (after the interface is implemented, please provide the interface name/interface address to the on-site implementation technicians in a timely manner, which needs to be configured to the ICS system)
- Provider: Third-party system
- Caller: ICS integrated control system
- Interface protocol: HTTP
- Function description: This interface is for ICS to proactively report the scanning information of the scanning door to the third-party system; it needs to be used in conjunction with the "task status reporting interface". If the status of the task status report passes through the scanning door and the third party has not received the interface information, the third-party system needs to determine that the task is a scanning abnormality.
- Usage scenario: The third-party system needs to know the scanning information of the scanning door when the AMR passes through the scanning door.
- Calling logic: After ICS obtains the scanning information of the scanning door, it reports to the third-party system without retrying.

3.2.6.2 Parameter details

Table 3-27 Actively report code reading information

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
Request	scanDoorData	shelfLayer	String	30	Yes	Number of shelf layers, default values are 1, 2, 3, 4 (default 1)
		locator	String	10	No	Cargo location number (provide when identifiable)
		boxNo	JSONArray	20	No	Code

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						information read by the scanning door
		extend	JSONObject	10	No	Extension fields
	doorCode		String	4	Yes	Scan the door number
	triggerTime		String	32	Yes	Scan code timestamp (ms value int64_t converted to string)
	orderId		String	10	No	Task order number
	shelfNumber		String	10	No	The shelf number that the latent lifting AMR carries through the code scanning door
Answer	code		int	6	Yes	Return status code: 1000: Success Other: Failed
	desc		string	64	Yes	Status code description

3.2.6.3 Examples

ask:

```
{
"scanDoorData": [{
"shelfLayer": "1",
"locator": "",
"boxNo": ["JA3C022763", "JA3C022516", "JA3C008691"]
```

```
}],
"doorCode": "S001",
"shelfNumber": "SZ0021",
"orderId": "RCS_1653241332"
}
```


Answer:


```
{
"code": 1000,
"desc": "Request successful"
}
```


3.2.7 Request to enter/leave transit area

3.2.7.1 Interface details

- Interface name: provided by the third-party system (after the interface is implemented, please provide the interface name/interface address to the on-site implementation technicians in a timely manner, which needs to be configured to the ICS system)
- Provider: Third-party system
- Caller: ICS integrated control system
- Interface protocol: HTTP
- Function description: This interface is for ICS to actively request the third-party system whether "AMR is allowed to enter/leave the passage area"; for example: the machine has a safety grating and needs to request to enter/leave the scene.
  - ◇ Request entry: The AMR arrives at the entrance point of the transit area and requests entry. The AMR enters after the third-party system returns permission for entry. If the third-party system returns that entry is not allowed, the AMR will stop at the entry point and apply until the third-party system allows entry. If the entry process changes from allowing entry to not allowing entry, the AMR will make an emergency stop.
  - ◇ Notification of entry: When the AMR arrives at the entrance point of the transit area, ICS notifies the third-party system that the AMR is ready to enter. The AMR enters directly without the need for the third-party system to return permission to enter.
  - ◇ Request to leave: When the AMR reaches the exit point of the route area, it will request to leave. The AMR will leave after the third-party system returns permission to leave. If the third-party system returns that it does not allow to leave, the AMR will stop at the exit point and request to leave until the third-party system allows to leave.
  - ◇ Notify departure: When the AMR arrives at the exit point of the transit area, ICS notifies the third-party system that the AMR is ready to leave. The AMR leaves directly without the need for the third-party system to return permission to leave.
- Usage scenario: Before the AMR enters a specific location (such as a machine or robotic arm), it needs to request permission from a third party to enter/leave before it can enter/leave the scene.

3.2.7.2 Parameter details

Table 3-28 Request to enter/leave transit area

Parameter type	Parameter name	Field Type	Maximum length	Is it required?	Description
Request	askZoneCode	string	100	Yes	Passing area number (machine/grating number)
	orderId	string	100	No	Third-party order

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						ID (only effective when there is a third-party system task on the current device)
	payload		int	2	No	The device's current no-load status: 1: Load 0: No load
	action		int	10	Yes	How to apply: 1: Request entry 2: Notify of departure 3: Notification entry 4: Request to leave
	deviceCode		string	64	Yes	Device serial number
	areaId		int	10	No	RegionId
Answer	code		int	6	Yes	Return status code: 1000: Success Other: Failed
	desc		string	64	Yes	Status code description
	data	enterStatus	int	2	No	Returns whether entry signal is allowed: 1: Entry allowed 0: No entry allowed Notification entry does not require returning this field
		leaveStatus	int	2	No	Returns whether the leave signal is allowed: 1: Allowed to leave 0: No leaving allowed Notify leaving without returning

Parameter type	Parameter name		Field Type	Maximum length	Is it required?	Description
						this field

3.2.7.3 Examples

ask:

```
{
"askZoneCode":"tujin1",
"areaId":11,
"action":1,
"deviceCode":"tuqs0004",
"orderId":"test001",
"payLoad":1
```

Answer:

```
{
"code": 1000,
"data":{
"enterStatus":1,
"leaveStatus":0
},
"desc": "Request successful"
}
```

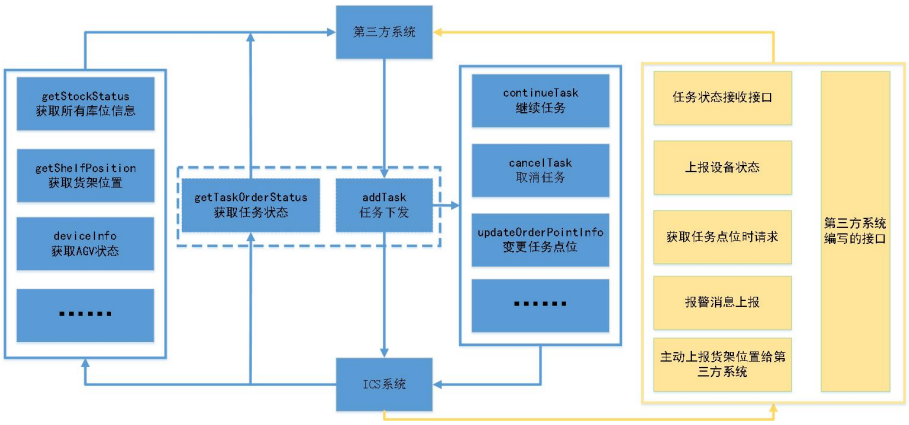
Chapter 4 Interaction Process

4.1 Basic principles

- For QR code navigation, the map is a collection of QR code values and coordinates. The QR code value can be named to indicate the actual application location. The naming supports English and numbers, for example:
  QR code value → named value
  →IRAYPLE00001
- For laser navigation, the map is a collection of nodes and coordinate values. The node can be named to indicate the actual application location. The naming supports English and numbers and the length supports 4-10 characters.
- Task template: AMR is a passive execution device. When executing a task, it performs actions according to the task template configuration section by section. The task template is used to configure the execution actions of these subtasks.
- Business template: It is a template compiled according to the business scenario of the project, that is, the business template is a collection of task templates, which includes the subtasks of the entire business. Therefore, modelProcessCode corresponds to the business template.
- For AMR transportation, the overall business is to move from one coordinate point to another, and then perform corresponding actions at the coordinate points (lifting equipment: lifting operation; roller equipment: transfer operation, etc.). The business template actually abstracts the AMR's transportation business into specific tasks. After the task is sent to the AMR, the AMR performs the corresponding action.

4.2 Interface calling relationship

Figure 4-1 Interface call relationship diagram



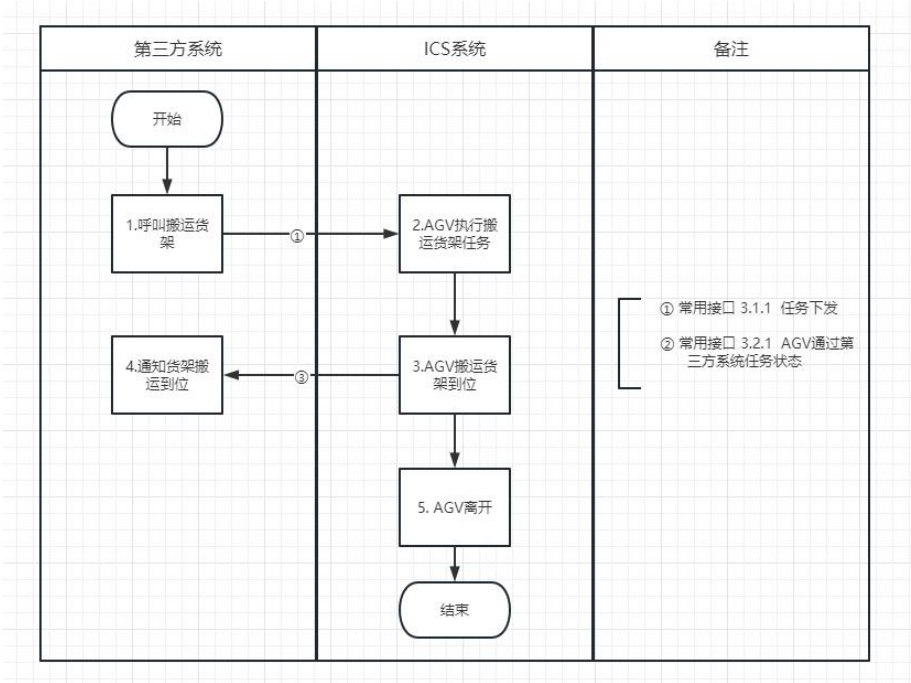
4.3 Interaction process diagram

Figure 4-2 Interaction process diagram

4.4 Typical scheduling scenarios

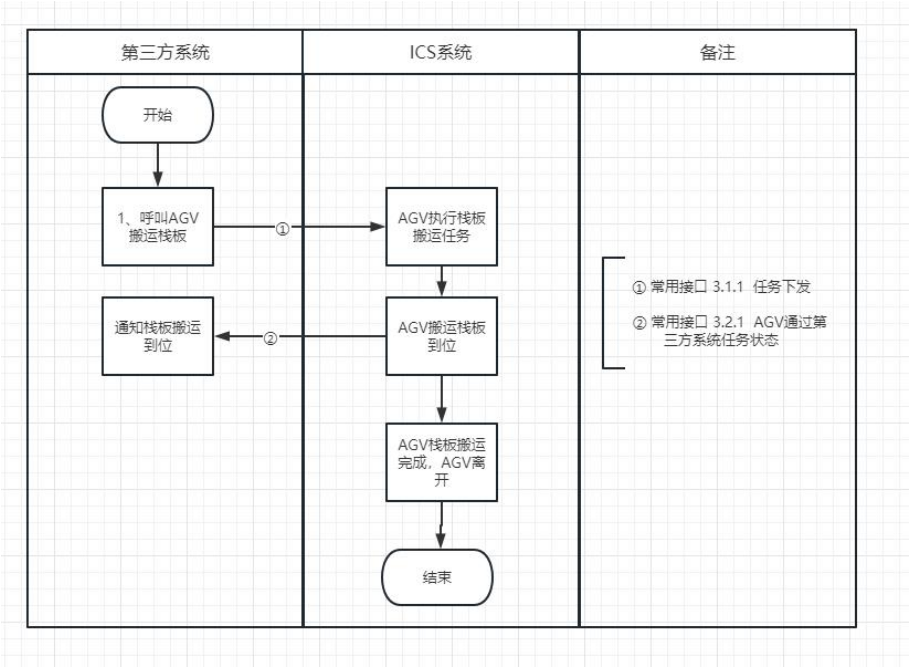
4.4.1 Ordinary shelf handling scenario

Figure 4-3 Ordinary shelf handling scene



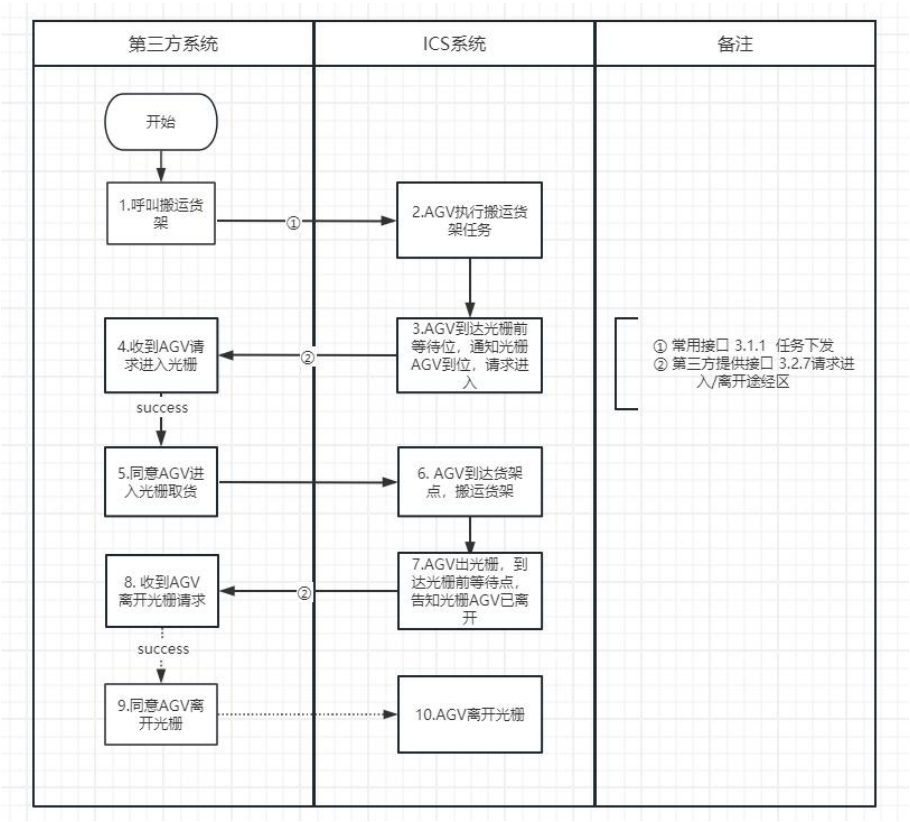
4.4.2 Ordinary forklift transporting pallets scenario

Figure 4-4 Scene of ordinary forklift transporting pallets



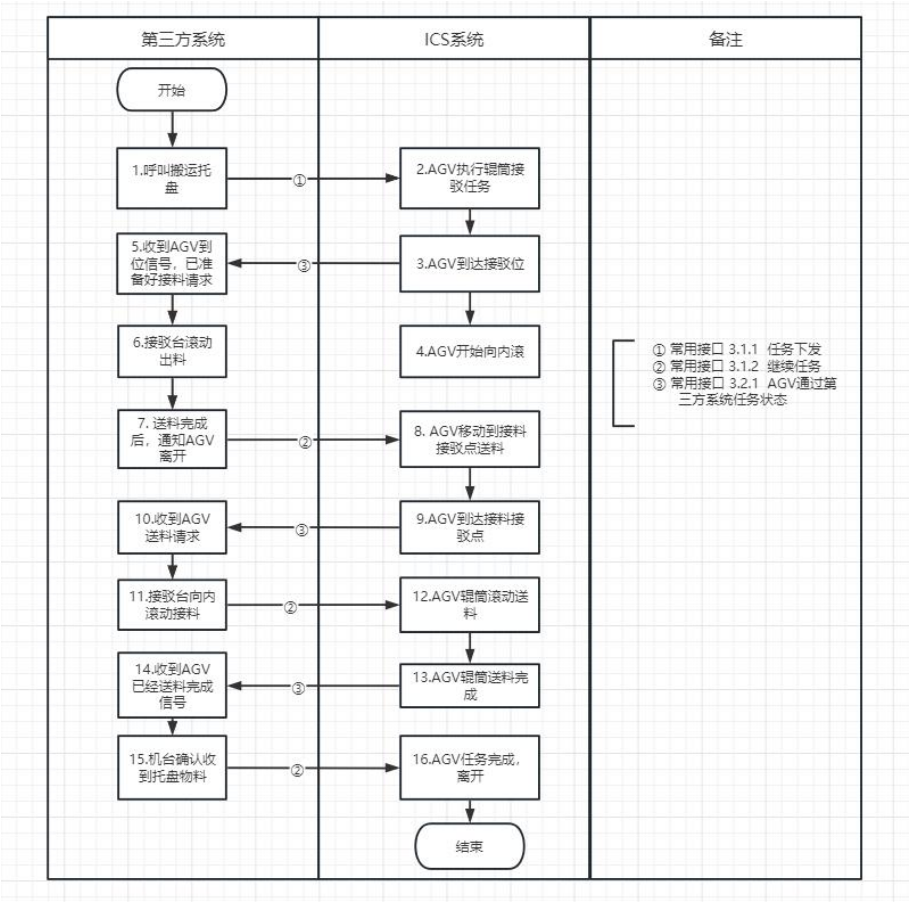
4.4.3 Transport scenarios in and out of gratings

Figure 4-5 Scene of transporting in and out of grating



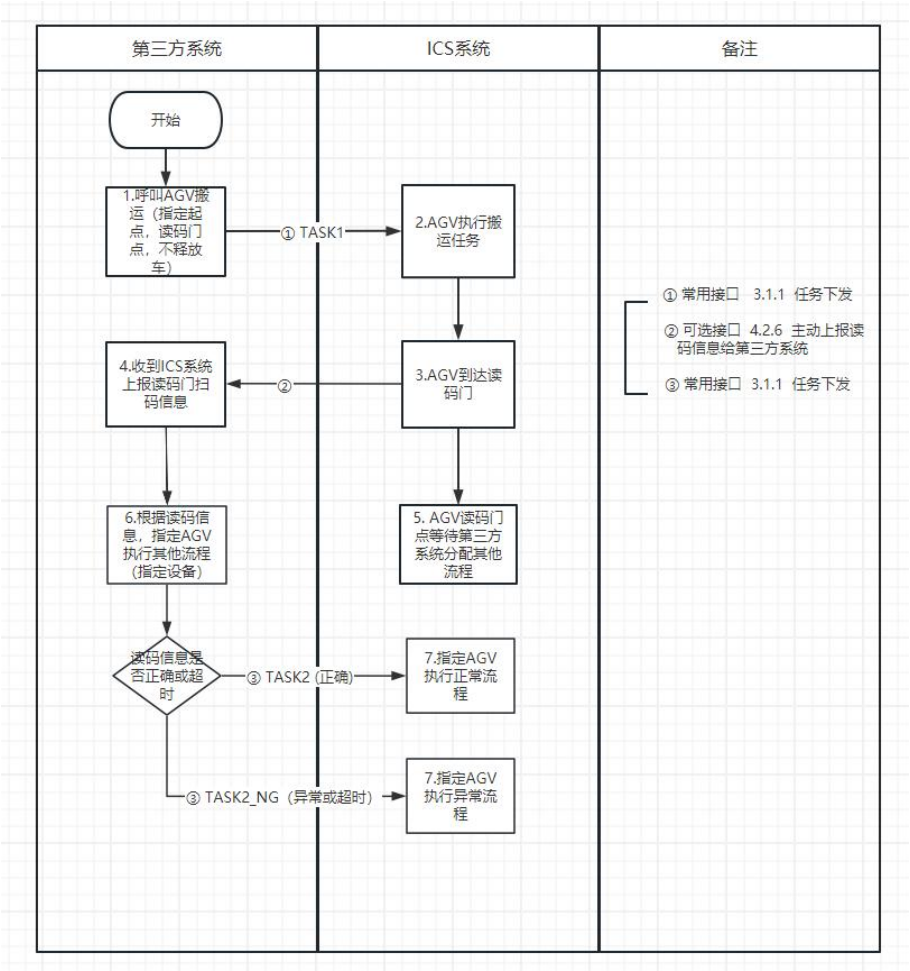
4.4.4 Roller truck transport scenario

Figure 4-6 Roller truck transport scene



4.4.5 AMR code scanning door scenario

Figure 4-7 AMR code scanning door scene



Chapter 5 Third-party system development recommendations

Call frequency description

Table 5-1 Call frequency

Serial number	Interface description	Call frequency
1	“2.1.1 Task dispatched"	Recommended frequency is less than 50 times/second
2	“2.1.2 Continue the task*”	Called when business needs
3	“2.1.3 Cancel a task*”	Called when business needs
4	“ Task status reporting interface"	If the third party does not return 1000 received, the default retry interval is 2 seconds and retries are made 5 times.
5	“3.1.1.1 Query task status"	It is recommended that the total request frequency be less than 50 times/second
6	“3.1.1.2 Change mission location"	Called when business needs
7	“3.1.1.3 Update task priorities"	Called when business needs
8	“3.1.1.4 End of mission	Called when business needs
9	“3.1.1.5 Pre-scheduling tasks	Called when business needs
10	“3.1.1.6 Rotating Shelves	Called when business needs
11	“3.1.1.7 Query task list"	Recommended frequency is less than 10 times/second
12	“3.1.2.1 GetAMRstate"	Recommended frequency is less than 10 times/second
13	“3.1.2.2 Get real-time equipment alarms and detailed planning information"	Recommended frequency is less than 10 times/second
14	“3.1.2.3 Get device task path"	Recommended frequency is less than 10 times/second
15	“3.1.2.4 Control device pause/recover"	Called when business needs
16	“3.1.2.5 Release device pre-emption	Called when business needs
17	“3.1.2.6 Send charging	Called when business needs
18	“3.1.2.7 Stop charging	Called when business needs
19	“3.1.3.1 Get shelf location"	Recommended frequency is less than 20 times/second
20	“3.1.3.2 Change Shelf Location"	Called when business needs

Serial number	Interface description	Call frequency
21	“3.1.3.3 Get all storage location information"	Recommended frequency is less than 20 times/second
22	“3.1.3.4 Change Location Status"	Called when business needs
23	“3.1.4.1 Apply to enter and exit the traffic control area"	Called when business needs
24	“3.1.4.2 Enable/"No-go zones"	Called when business needs
25	“3.1.5 Get map interface"	Called when business needs
26	“3.2.1 Report device status to third-party system"	Polling report, default is 2 seconds/time, the interface can be configured and modified
27	“3.2.2 Request a third-party system to obtain the mission point during the mission"	When a subsequent target point is required in a task and the third-party system has not returned, the default interval for obtaining it is 2 seconds.
28	“3.2.3 "Alarm message reporting"	Called when an AMR alarm occurs
29	“3.2.4 Actively report shelf location to third-party systems"	Polling report, default is 2 seconds/time, the interface can be configured and modified
30	“3.2.5 Actively report the status of the charging pile to the third-party system"	Polling report, default is 2 seconds/time, the interface can be configured and modified
31	“3.2.6 Actively report the scanning door scanning information to the third-party system"	Called when the scanning door scans the code
32	“3.2.7 Request access/Leave the transit area	Polling request, default is 2 seconds/time, the frequency can be configured and modified

Appendix 1 Equipment alarm type comparison table

Appendix Table 1-1 Equipment alarm type comparison table

Alarm Type	Whether to support ending alarm	Alarm type description
1	Support	Device abnormally offline
2	Support	Equipment failure
3	Support	Low battery alarm
4	Support	AMR equipment abnormality
5	Not supported	Device upgrade exception
6	Not supported	Node failure
8	Support	Navigation anomaly
11	Not supported	Map update failed
12	Support	Equipment failure alarm
15	Not supported	High temperature
20	Support	Charging station abnormality
21	Support	Charging abnormality
22	Support	Shelf abnormality
23	Support	Path exception
24	Not supported	Configuration exception
25	Support	Load abnormality
26	Support	Abnormal storage location
28	Support	Equipment collision
29	Not supported	Laser sensor abnormality
30	Not supported	Cargo overheight detection abnormality
31	Support	Device replacement abnormality
40	Not supported	RFID abnormality
41	Support	Access control abnormality
42	Support	Elevator control abnormality
43	Support	Storage location sensor abnormality
44	Support	Smart camera abnormality
45	Not supported	Waiting for confirmation
46	Support	Service exception
48	Support	Pager abnormality
49	Support	Industrial indicator light abnormality
50	Exceptions are not supported	Identifying anomalies

Appendix 2 Subtask type comparison table

Appendix Table 2-1 Subtask type comparison table

Subtask Type	Subtask type description
1	Empty car moving
2	Mobile racks/pallets (load arrival)
3	Roller rolling (loading)
4	Lifting rack
5	Put down the shelves
10	Moving Goods
12	Flip/Reset
13	Empty vehicle movement (to identification preparation point, no height adjustment)
14	Roller rolling (unloading)
16	Forklift cargo box
17	Put down the cargo box
18	Hook
19	Decoupling
20	Empty vehicle movement (to identification preparation point, including height adjustment)
21	Pallet docking (stacking forklift AMR)
22	Moving pallets (including: lifting while moving, lowering while moving)
23	Putting down the pallet (pallet handling AMR)
24	Put down the pallet (stacking forklift AMR)
26	Put down the tines
28	Pallet docking (pallet handling AMR)
29	Shelf identification
31	Moving pallets (lifting while moving)
32	Moving pallets (descending as you go)
33	Mobile shelves/pallets/boxes (equipment arrival)
37	Fork tine height adjustment
38	Roller rolling (roller has no sensor)
43	Placing bins/pallets
45	Roller rolling (sorting and loading)
46	Roller rolling (sorting and unloading)
47	Manual intervention (sorting and loading)
48	Manual intervention (sorting and unloading)

Subtask Type	Subtask type description
49	Lifting pallets (pallet handling AMR)
57	Placing pallets (stacking forklift AMR)
58	Placing material boxes (coaxial forklift AMR)
59	Moving empty vehicle (picking up material boxes while moving)
62	Towing hook
63	Towing unhooking
66	Placing boxes/pallets V2
68	Multi-model pallet docking (stacking forklift AMR)
69	Multi-model pallet docking (pallet handling AMR)
70	Multi-model shelf recognition

Appendix 3 Legal Statement

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About this manual

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Appendix 4 Safety Commitments and Recommendations

The product adheres to the principle of minimum necessary data collection and service minimization, strictly prohibits the implantation of backdoors, removes unnecessary and unsafe services (such as Telnet, etc.), and continuously introduces innovative security technologies, striving to enhance and improve the security capabilities of the product.
Software platform security not only requires manufacturers' continuous attention and efforts in R&D, production, and delivery, but also requires users' active participation, continuous attention and improvement of the environment and methods of platform use, to better ensure the safe operation of the software platform after it is put into use. To this end, users are advised to use the software platform safely, including but not limited to:

Account Management

- The length must not be less than 8 characters.
- Contain at least two character types, including uppercase and lowercase letters, numbers, and symbols.
- Does not contain the account name or the account name in reverse order.
- Do not use consecutive characters such as 123, abc, etc.
- Do not use overlapping characters, such as 111, aaa, etc.

1. Change your password regularly
   It is recommended that you change your account password regularly to reduce the risk of being guessed or cracked.
2. Reasonably allocate accounts and permissions
   Add new accounts appropriately based on business and management needs, and assign them the minimum set of permissions.
3. Enable account lock
   The account lock function is enabled by factory default. It is recommended that you keep it enabled to protect account security. After the attacker fails to enter the password multiple times, the corresponding account and source IP will be locked.
4. Set and update password reset information in a timely manner
   The software platform supports the password reset function. In order to reduce the risk of this function being exploited by attackers, please set the password reset related information in a timely manner. If any information changes, please modify it in time. When setting security questions, it is recommended not to use answers that are easy to guess, and to set different answers for different security questions.
5. Enable account binding IP/MAC
   To further improve security, it is recommended that you bind your account to the terminal IP/MAC where the commonly used client is located.

Service configuration

1. Enable HTTPS

   It is recommended that you enable HTTPS and access WEB services through a secure channel.

2. Turn off non-essential services and use safe mode

   If there is no need, it is recommended that you turn off SSH, SMTP and other functions to reduce the attack surface. If necessary, it is strongly recommended that you use a safe mode, including but not limited to:

   ● SMTP: Select TLS to access the email server.

   ● FTP: Select SFTP and set a complex password.

Network configuration

1. Configure firewall whitelist

   It is recommended that you configure the firewall whitelist function and only allow IPs on the whitelist to access the software platform. Therefore, please be sure to add your computer IP address and the corresponding software platform IP address to the whitelist.

2. Network segmentation/isolation

   It is recommended that you divide the video surveillance network and the office network into different VLANs on switches and routers to isolate the two networks and prevent attackers from invading the office network and then using it as a springboard to attack the video surveillance network.

Security Audit

1. View online users

   It is recommended that you check online users from time to time to identify whether there are any illegal users logging in.

2. Check the platform log

   By checking the logs, you can obtain the IP information of the person attempting to log into the software platform, as well as the operation information of the logged-in user.

Physical protection

It is recommended that you provide physical protection for the software platform, such as placing the software platform in a dedicated computer room or cabinet, and managing access control and keys to prevent unauthorized personnel from damaging the hardware or making physical contact.

Border security configuration

It is recommended that you deploy security products at the boundary of the software platform and take necessary authorized access, access control, intrusion prevention and other measures to implement security protection within the software platform.

Appendix 5 Revision History

Appendix Table 5-1 Revision Record

Version number	Software version number	Revisions	Release Date
V3.2.0	ICS3.5&RCS4.5 and later versions	● When issuing a task or CTU task, use the shelfNumber field as the container number. ● Add subOrderId as the request identifier in the continue task and cancel task interfaces ● When reporting the task status, modify the shelfNumber description to support its use as the material box number transported by the CTU. Add hasContainer as a description of whether the container is scanned. ● Storage location status changes, supports transmitting extended information when the storage location is full ● Modify the URL address of the interface for applying to enter the traffic control area ● Added CTU to actively report code reading information to third-party systems ● Added a new safety interaction interface for transit areas ● Suggestions to increase calling frequency of some interfaces ● Revised some field descriptions	2024.12
V3.1.0	ICS3.4&RCS4.4 and later versions	Add device pause mode	2024.07
V3.0.8	ICS3.3&RCS4.3 and later versions	●Task issuing interface ◇ Delete the loadType field ◇ Add the shelfModel field to view the description of the custom model ●Query task status interface The subTaskStatus field only retains one description ●End task interface Add description of interface scope ● Pre-scheduling tasks The taskTyp field value needs to be a single empty car moving task process template number ● Select shelf interface	2024.05

Version number	Software version number	Revisions	Release Date
		Add A, B configuration description ● Get AGV status Modify the error description of deviceCode ● Control device pause/resume Add stopType description, currently not supported ● Get device task path ◇ fatherAreaId, isCross fields are deprecated ◇ DeviceCode adds support for multiple descriptions ● Report device status to third-party system Added instructions for device status reporting in RCS client configuration	
V3.0.7	ICS3.3&RCS4.3 and later versions	● Add active reporting of charging pile status ● Added an overview of the code reading gate process scenario and a code reading information reporting interface ● Added new charging interface ● Added a new stop charging interface ● Control device pause/resume to add task order number: orderId field ● When issuing tasks, add extraInfo field in task reporting ● The device pause interface adds a device stop type field ● The alarm message reporting interface adds an alarm status field	2024.01
V3.0.6	ICS3.2&RCS4.2 and later versions	● Add interaction process diagrams and scenario descriptions ●Add task list acquisition interface ●Add map data acquisition interface ● Add order number in the interface for obtaining device information ● Add enumeration values for task status reporting: Pickup completed, Delivery completed ● Actively obtain task status and add subtask related fields ● Equipment status reporting and equipment status acquisition include adding task order number and shelf number	2023.05
V3.0.5	ICS3.1&RCS4.1	Streamline task delivery parameters and remove	2023.04

Version number	Software version number	Revisions	Release Date
	and later versions	some non-essential parameters	
V3.0.4	ICS3.1&RCS4.1 and later versions	●Add interface for applying for traffic control area ● Remove the temporary task sending interface ● Modify some descriptions ● Add enable and disable functions for restricted areas ●Add pre-scheduling task distribution ● Add interface for setting pre-scheduling parameters	2023.03
V3.0.3	ICS3.1&RCS4.1 and later versions	New multi-vendor AGV traffic control interface	2022.12
V3.0.2	ICS3.0&RCS4.0 and later versions	Added appendix subtask type comparison table	2022.09
V3.0.1	ICS2.0&RCS3.0 and later versions	● Delete device-related safety instructions ● Revise the error code comparison table and add an alarm message type comparison appendix ●	2022.08
V2.1.2	ICS2.0&RCS3.0 and later versions	● Changes in the interface address for rotating shelves, updating task priorities, and obtaining storage location status ● Add description of obtaining algorithm planning status and usage methods	2021.12
V2.1.1	ICS2.0&RCS3.0 and later versions	● Added introduction to interaction process ● Modify some interface descriptions in the document and add interface call request examples ● Add interfaces such as “change storage location status” and “obtain equipment algorithm planning status”	2021.12
V2.1.0	ICS1.0&RCS2.4 and later versions	Change the English font (including page numbers) to "Myriad Pro"	2020.12
V2.0.0	ICS1.0&RCS2.4 and later versions	● Delete the instructions related to “Enable whitelist” and “Connect devices using PoE” ● Added new instructions in “Building a secure network environment”: “Turn on device	2020.08

Version number	Software version number	Revisions	Release Date
		IP/MAC address filtering to limit the range of hosts allowed to access the device”	
V1.0.0	ICS1.0&RCS2.4 and later versions	First release	2019.08